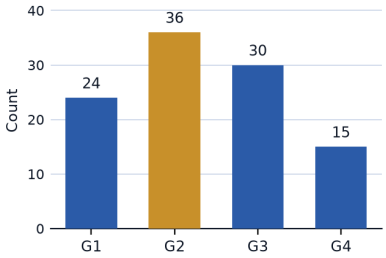
Sequences & Series

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Box plots and measures of spread

For the model $f(x) = 4x + 4$ in box plots and measures of spread, evaluate $f(5)$.

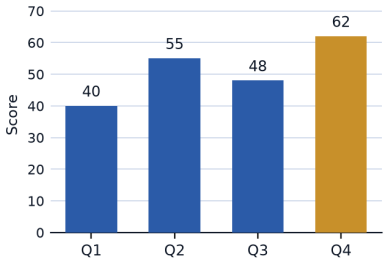


2 [Foundational] • Target: Box plots and measures of spread

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

3 [Foundational] • Target: Data distributions

For the model $f(x) = 5x + 5$ in data distributions, evaluate $f(2)$.



4 [Foundational] • Target: Data distributions

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

5 [Foundational] • Target: Data distributions

A data set recorded for data distributions consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean? Show your complete algebraic work.

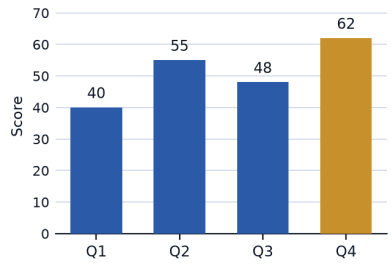
6 [Foundational] • Target: Data distributions

For the model $f(x) = 5x + 5$ in data distributions, evaluate $f(3)$.

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7 [Foundational] • Target: Data distributions

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Data distributions

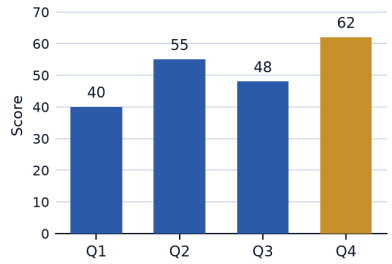
For the model $f(x) = 5x + 5$ in data distributions, evaluate $f(4)$.

9 [Intermediate] • Target: Data distributions

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

10 [Intermediate] • Target: Data distributions

For the model $f(x) = 5x + 5$ in data distributions, evaluate $f(5)$.

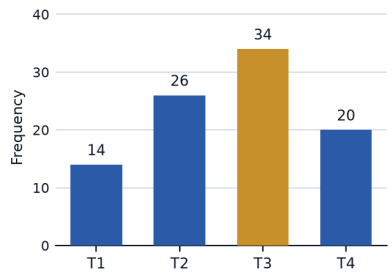


11 [Intermediate] • Target: Data distributions

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

12 [Intermediate] • Target: Comparisons of data sets

For the model $f(x) = 6x + 6$ in comparisons of data sets, evaluate $f(2)$.



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13 [Intermediate] • Target: Comparisons of data sets

A data set recorded for comparisons of data sets consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

14 [Intermediate] • Target: Comparisons of data sets

For the model $f(x) = 6x + 6$ in comparisons of data sets, evaluate $f(3)$.

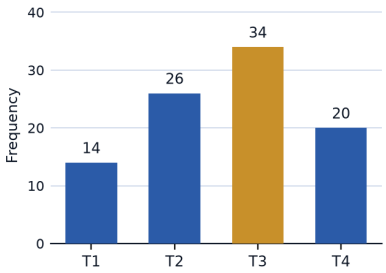
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Comparisons of data sets • Cross-Lesson: 11.01 Statistics & Data Analysis

For the model $f(x) = 6x + 6$ in comparisons of data sets, evaluate $f(4)$.

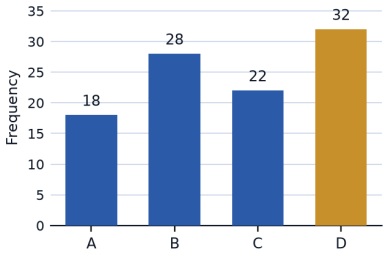
16 [Challenging] • Target: Comparisons of data sets • Cross-Lesson: 2.01 Quadratic Functions

For the model $f(x) = 6x + 6$ in comparisons of data sets, evaluate $f(5)$.



17 [Challenging] • Target: Two-way frequency tables • Cross-Lesson: 3.01 Exponents & Polynomials

For the model $f(x) = 2x + 7$ in two-way frequency tables, evaluate $f(2)$.



18 [Challenging] • Target: Two-way frequency tables • Cross-Lesson: 4.01 Functions & Transformations

A data set recorded for two-way frequency tables consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

19 [Challenging] • Target: Two-way frequency tables • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

For the model $f(x) = 2x + 7$ in two-way frequency tables, evaluate $f(3)$.

20 [Challenging] • Target: Two-way frequency tables • Cross-Lesson: 6.01 Systems & Inequalities

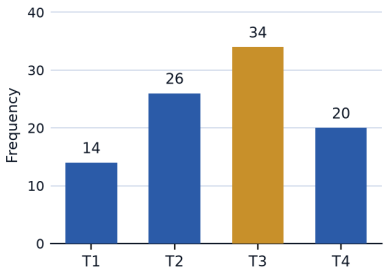
For the model $f(x) = 2x + 7$ in two-way frequency tables, evaluate $f(4)$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Frequency data displays

For the model $f(x) = 2x + 9$ in frequency data displays, evaluate $f(2)$.



a. 13

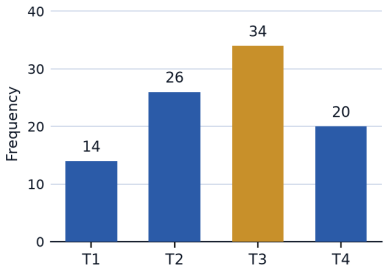
b. 15

c. 4

d. 17

22 [Foundational] • Target: Frequency data displays

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.



a. $x = 2$ and $x = 8$

b. $x = 2$ and $x = 7$

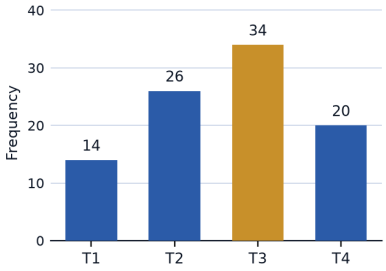
c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

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23 [Foundational] • Target: Frequency data displays

For the model $f(x) = 2x + 9$ in frequency data displays, evaluate $f(5)$.



a. 19

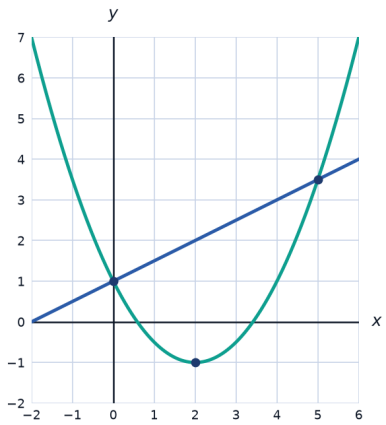
b. 21

c. 10

d. 23

24 [Foundational] • Target: Measures of center

For the model $f(x) = 3x + 3$ in measures of center, evaluate $f(2)$.



a. 9

b. 12

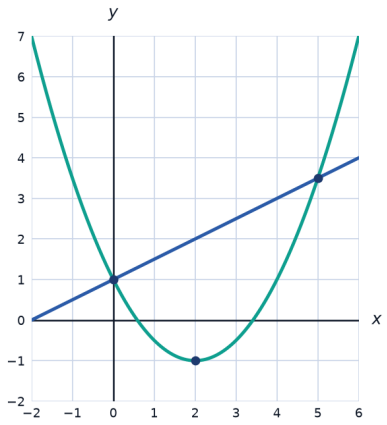
c. 6

d. 13

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25 [Foundational] • Target: Measures of center

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

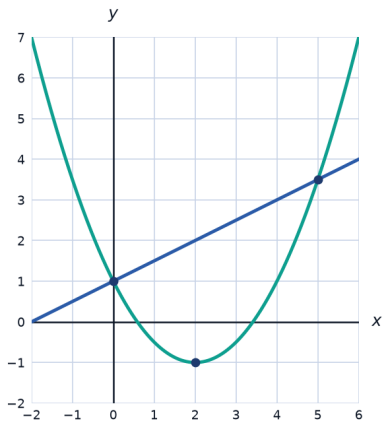
b. $x = 2$ and $x = 4$

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

26 [Foundational] • Target: Measures of center

For the model $f(x) = 3x + 3$ in measures of center, evaluate $f(5)$.



a. 18

b. 21

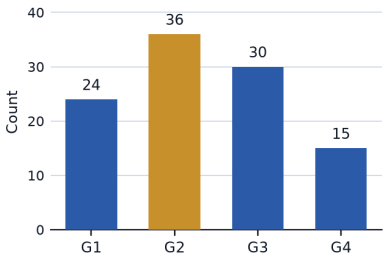
c. 15

d. 22

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27 [Foundational] • Target: Box plots and measures of spread

For the model $f(x) = 4x + 4$ in box plots and measures of spread, evaluate $f(2)$.



a. 12

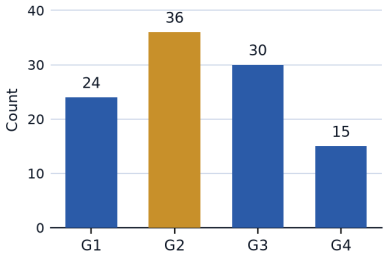
b. 16

c. 8

d. 16

28 [Foundational] • Target: Box plots and measures of spread

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

b. $x = 2$ and $x = 5$

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Frequency data displays

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

a. $x = 1$ and $x = 7$

b. $x = 1$ and $x = 6$

c. $x = -1$ and $x = -6$

d. $x = 7$ and $x = 6$

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30 [Intermediate] • Target: Frequency data displays

A data set recorded for frequency data displays consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

31 [Intermediate] • Target: Frequency data displays

For the model $f(x) = 2x + 9$ in frequency data displays, evaluate $f(3)$.

a. 15

b. 17

c. 6

d. 19

32 [Intermediate] • Target: Frequency data displays

For the model $f(x) = 2x + 9$ in frequency data displays, evaluate $f(4)$.

a. 17

b. 19

c. 8

d. 21

33 [Intermediate] • Target: Frequency data displays

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

34 [Intermediate] • Target: Frequency data displays

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

a. $x = 4$ and $x = 10$

b. $x = 4$ and $x = 9$

c. $x = -4$ and $x = -9$

d. $x = 13$ and $x = 36$

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35 [Intermediate] • Target: Measures of center

A data set recorded for measures of center consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

36 [Intermediate] • Target: Measures of center

For the model $f(x) = 3x + 3$ in measures of center, evaluate $f(3)$.

a. 12

b. 15

c. 9

d. 16

37 [Intermediate] • Target: Measures of center

For the model $f(x) = 3x + 3$ in measures of center, evaluate $f(4)$.

a. 15

b. 18

c. 12

d. 19

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Measures of center • Cross-Lesson: 4.01 Functions & Transformations

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

39 [Challenging] • Target: Measures of center • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

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40 [Challenging] • Target: Box plots and measures of spread • Cross-Lesson: 6.01 Systems & Inequalities

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ and $x = 4$

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

41 [Challenging] • Target: Box plots and measures of spread • Cross-Lesson: 8.01 Factoring Polynomials

For the model $f(x) = 4x + 4$ in box plots and measures of spread, evaluate $f(3)$.

a. 16

b. 20

c. 12

d. 20

42 [Challenging] • Target: Box plots and measures of spread • Cross-Lesson: 9.01 Radicals & Exponents

For the model $f(x) = 4x + 4$ in box plots and measures of spread, evaluate $f(4)$.

a. 20

b. 24

c. 16

d. 24

43 [Challenging] • Target: Box plots and measures of spread • Cross-Lesson: 1.01 Linear Equations & Lines

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

44 [Challenging] • Target: Data distributions • Cross-Lesson: 10.01 Rational Expressions

A data set recorded for data distributions consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

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45 [Challenging] • Target: Comparisons of data sets • Cross-Lesson: 11.01 Statistics & Data Analysis

A data set recorded for comparisons of data sets consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5

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